

Swarup Mahapatra

LEAD SOFTWARE ENGINEER

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SUMMARY

Engineering leader with nearly 13 years of experience defining technical strategy, shaping architecture, and delivering resilient distributed systems across fintech, healthcare, edtech, and enterprise software. Leads ambiguous initiatives from discovery through production, aligns product and engineering decisions across teams, and builds cloud-native platforms that improve operational resilience, delivery velocity, and software quality.

PROFESSIONAL EXPERIENCE



Lead Software Engineer

Everest Engineering, Bengaluru, India | June 2022 - Present

- Set technical direction across multiple client engagements, guiding teams of 3-6 engineers from problem framing and architectural discovery through production delivery.
- Influenced product and business stakeholders on technical strategy, architecture trade-offs, sequencing, and investment decisions in complex, high-ambiguity environments.
- Established shared engineering practices for observability, distributed tracing, cloud-native reliability, Domain-Driven Design, clean architecture, and delivery quality.
- Mentored engineers and raised team capability through system design reviews, architectural guidance, and hands-on coaching across client programs.

PROJECTS

Clinical Research Data Platform | Confidential healthcare client

October 2025 - Present

- Led a team of seven engineers and owned the technical strategy, architecture, and delivery roadmap for a clinical research data platform.
- Defined ingestion and preprocessing architecture to transform raw clinical study data from multiple sources into a standardized schema in an AWS data lake.
- Generated data-mapping specifications with Claude Sonnet 4.6 and built Python and AWS workflows using S3, Lambda, DynamoDB, Step Functions, and Glue to support allergy-study analysis across ~10 studies per year and ~10,000 participants per study.
- Achieved ~80% parsing accuracy and reduced manual review effort by ~90% through automated classification of consent and sensitive information.

Python

AWS S3

AWS Lambda

AWS DynamoDB

AWS Step Functions

AWS Glue

Claude Sonnet 4.6

Learning Management Platform | Confidential Australia-based edtech client

March 2025 - July 2025

- Led discovery, architectural definition, and end-to-end product delivery from inception for a custom learning management platform supporting personalized learning pathways.
- Set technical direction and coordinated a team of three engineers across frontend, backend, and infrastructure delivery.
- Built the backend with NestJS and Postgres, the frontend with Next.js, and provisioned AWS services with Terraform while using Sanity CMS for course content management.

TypeScript

NestJS

Next.js

Postgres

AWS

Terraform

Sanity CMS

Ethical Investment Platform | Confidential Australia-based fintech client

December 2023 - October 2024

- Owned product and technical delivery from inception through beta launch for an ethical investing and shareholder advocacy platform.
- Defined the architecture for a serverless trading platform using AWS Lambda, S3, API Gateway, SNS, SQS, DynamoDB, and MongoDB to support campaign pledges and progress tracking.
- Directed vendor communication and integrations across KYC, trading, and fund management platforms, applying Domain-Driven Design to establish scalable bounded contexts and service boundaries.

Node.js

DynamoDB

MongoDB

AWS Lambda

S3

API Gateway

SNS

SQS

Serverless

Education Platform Modernization | Confidential Australia-based edtech client

July 2022 - November 2023

- Maintained and upgraded public-facing APIs by introducing RESTful patterns and practices across legacy services.
- Drove adoption of Clean Architecture in selected legacy projects and redesigned the CI and deployment process, reducing deployment time from hours to minutes through Trunk-Based Development.
- Managed a team of five engineers and established a stronger engineering practice through Domain-Driven Design, architectural principles, and structured technical coaching.

Node.js

Postgres

REST APIs

CI/CD

Trunk-Based Development

DDD

Also used across engagements:

Python

TypeScript

NodeJS

Postgres

DynamoDB

Serverless

Docker

React

AWS

DDD



Senior Software Engineer

SOCASH PTE LTD (acquired by Nium), Bengaluru, India | November 2018 - May 2022

- Shaped the architecture and technical roadmap for a fintech payment platform, balancing reliability, throughput, extensibility, and operational resilience in a high-stakes transaction environment.
- Led integrations with banks and retail partners, expanding the merchant POS ecosystem through secure, partner-facing APIs across a multi-country payment footprint.
- Built and operated a payment gateway integrating 15 banks and vendors, supporting ~10,000 transactions per day for ~500 merchants across 4 countries.
- Re-architected a Node.js monolith into modular Java payment services, establishing clearer service boundaries, ownership models, and a platform-oriented design for long-term scalability.
- Automated provider failover and configuration management to detect downstream outages and switch payment routes, reducing manual intervention during partner disruptions and improving resilience.
- Built and scaled quality engineering practices with performance testing, monitoring, and observability for critical payment APIs, increasing system stability and operational visibility.

Java

NodeJS

Ruby (Test Automation)

AWS

Terraform

Postgres

Redis

RethinkDB (NoSQL DB)

DDD

TDD

CI using Jenkins



Software Engineer

Oracle, Bengaluru, India | March 2017 - October 2018

- Continued working on the Aconex product within Oracle for several months following the acquisition, supporting a smooth product and engineering transition.
- Designed and delivered backend capabilities for document review workflows of Aconex product, with direct ownership of customer-facing service behavior.
- Operated and deployed distributed microservices across regions, prioritizing availability, resilience, and predictable releases.
- Built REST APIs for real-time PDF review collaboration and document annotation workflows.
- Established automated UI testing infrastructure that improved regression coverage for core review features.

Java

Ruby (Test Automation)

AWS

Terraform

Postgres

MSSQL

DDD

TDD

CI using Jenkins



Software Consultant

Thoughtworks, Bengaluru, India | November 2015 - February 2017

- Delivered software for Bahmni, an open-source healthcare workflow platform, working across product, engineering, and implementation teams.
- Built API automation and performance testing capabilities that strengthened release quality and system feedback loops.
- Supported implementations for Médecins Sans Frontières (MSF), contributing to reliable workflows in mission-critical healthcare settings.

Selenium

Ruby (Test Automation)

Capybara

Gatling



Data Specialist

IBM, Bengaluru, India | November 2013 - October 2015

- Contributed to a big data credit risk platform, improving confidence in data quality and pipeline reliability.
- Analyzed source data and validated ETL mappings against business and technical requirements.
- Developed Python tooling to generate reliable ETL test data and streamline data preparation workflows.

Python

Hadoop

Hive

HBase

Pig

KEY SKILLS

Languages: Java, JavaScript, Ruby, Python

Databases: Postgres, DynamoDB

Cloud Technologies: Amazon Web Services

CI/CD: Jenkins, GitHub Actions, GitLab, Git

Infrastructure: Terraform, AWS CDK, Ansible, Docker

Practices: DDD, TDD, OOP, Functional Programming, System Design, Distributed Systems, Agile, Scrum

AI Tools: Prompt Engineering, RAG, Vector Databases, Spec Driven Development

EDUCATION

Bachelor of Technology (BTech) in Electronics and Communication Engineering

National Institute of Technology, Rourkela | July 2009 - May 2013

CGPA: 8.20